Improving the Efficiency and Effectiveness of LLM Knowledge Distillation for Conversational Search

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Introduction

Conversational Search (CS) considers retrieval of relevant documents based on conversational context. While Large Language Models (LLMs) significantly enhance query rewriting, their inference costs pose major efficiency challenges. Knowledge distillation (KD) balances effectiveness and efficiency by training a sparse retriever (Student) to mimic LLM query representations (Teacher). Recent state-of-the-art models like DiSCo use Kullback-Leibler Divergence (KLD) rather than Mean-Squared Error (MSE) for this distillation. Despite these gains, several mechanisms of KLD-based distillation remain understudied. We expand this research by investigating:

1. To what extent can a KL divergence distillation objective be combined with a contrastive InfoNCE for improved performance in conversational search?
2. How does the balance of positive and negative samples used in the KL Divergence objective affect distillation behavior and downstream retrieval performance?
3. How does increasing regularization in DiSCo influence the trade-off between sparsity, inference efficiency, and retrieval effectiveness, particularly for longer conversational queries?

Dual Loss

The standard KLD approach aligns distributions but lacks a fine-grained contrastive incentive. We propose a weighted combination of the KLD distillation loss and a contrastive InfoNCE loss:

$$Loss = \lambda \cdot InfoNCE_{sim} + (1 - \lambda) \cdot KLD$$

Here, $\lambda \in [0, 1]$ is a parameter to balance the contrastive and distillation incentives.

Table 1. Results on TopiOCQA after adding a weighted contrastive loss (InfoNCE) to the KL divergence. The weight of DiSCo and InfoNCE sum to 1. $\star$ is for significantly better and † for significantly worse results under a Bonferroni-corrected paired t-test with a 95% confidence level.

Loss	MRR	R@10	R@100	nDCG@3
DiSCo	0.409	0.676	**0.879**	0.396
DiSCo + 0.05 InfoNCE	0.414	**0.687**	0.875	0.400
DiSCo + 0.10 InfoNCE	0.421*	0.673	0.877	0.408*
DiSCo + 0.20 InfoNCE	**0.424***	0.676	0.874	**0.412***

The addition of a ranking loss improves the ranking capacity, increasing performance in terms of MRR and nDCG@3. Additionally, we observe that this can lead to higher Recall@k scores for lower k values.

Negative Sampling in the KL Divergence

Theoretically, drawing more samples from the teacher should improve the approximation of the KLD objective. However, in retrieval, the single positive document must remain sufficiently represented to provide a strong learning signal.

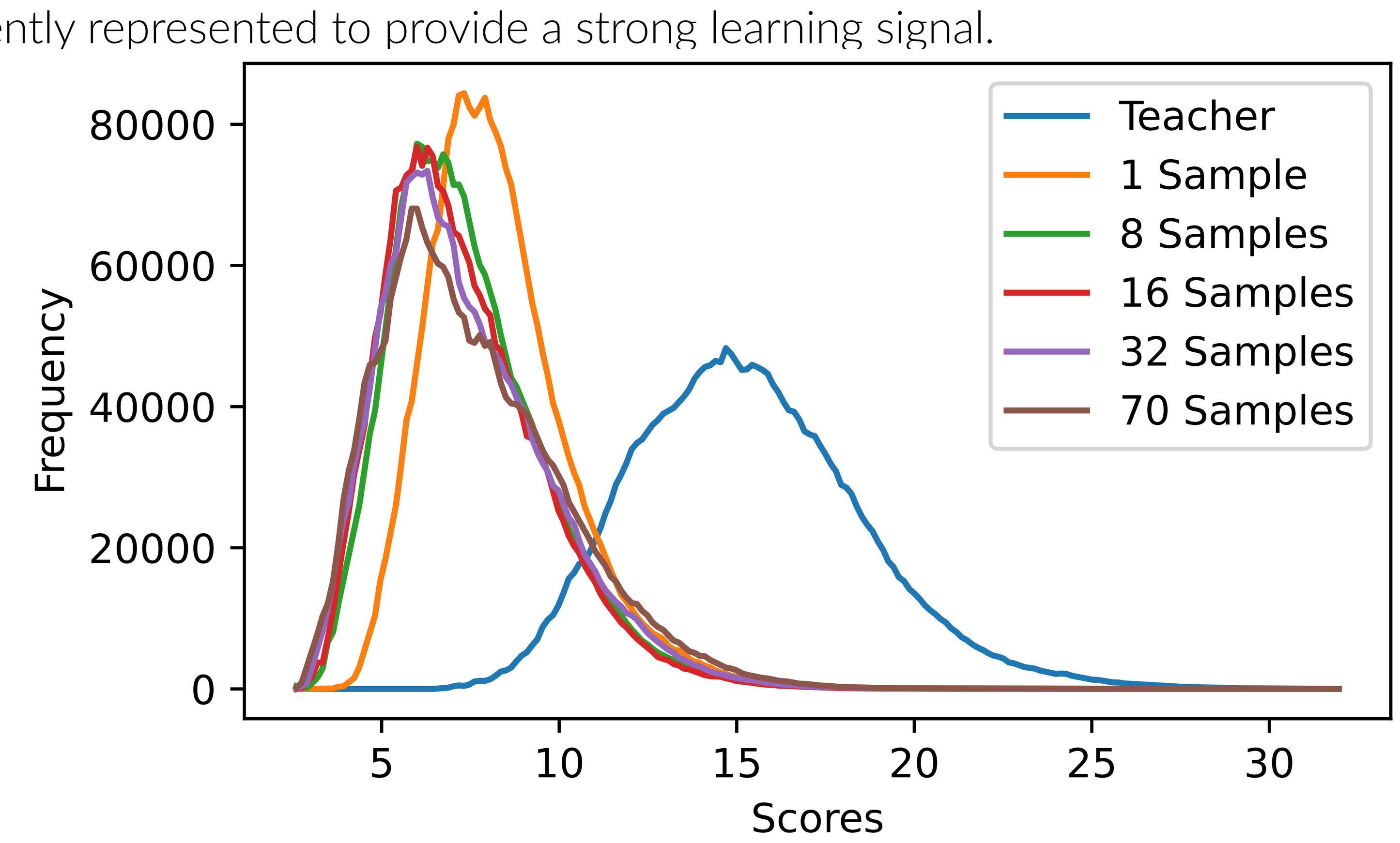

Figure 1. Predicted similarity-score distributions on TopiOCQA for student models trained with different numbers of KLD negatives, with the SPLADE MistralQR teacher for reference.

Table 2. TopiOCQA results for different numbers of KLD negatives. Each setting uses one positive sample.

Neg.	MRR	R@10	R@100	nDCG@3	D_{KL}
1	0.348†	0.613†	0.859†	0.326†	3.501
8	0.394†	0.657†	0.874	0.381†	3.861
16	0.409	**0.676**	**0.879**	**0.396**	3.698
32	**0.410**	0.662†	0.874	0.395	3.592
70	0.395†	0.655†	0.860†	0.381†	**3.221**

Although more negatives reduce the final KLD value, they do not improve retrieval performance. The best effectiveness is obtained around 16 negatives, suggesting that too many negatives dilute the influence of the positive sample.

Increasing Sparsity

Because the KLD objective is a relaxation compared to MSE-based methods, it should allow for more general representations. We apply increasing amounts of regularization (L_1 on query representations and FLOPS on document representations) to test if we can force sparsity without diminishing returns.

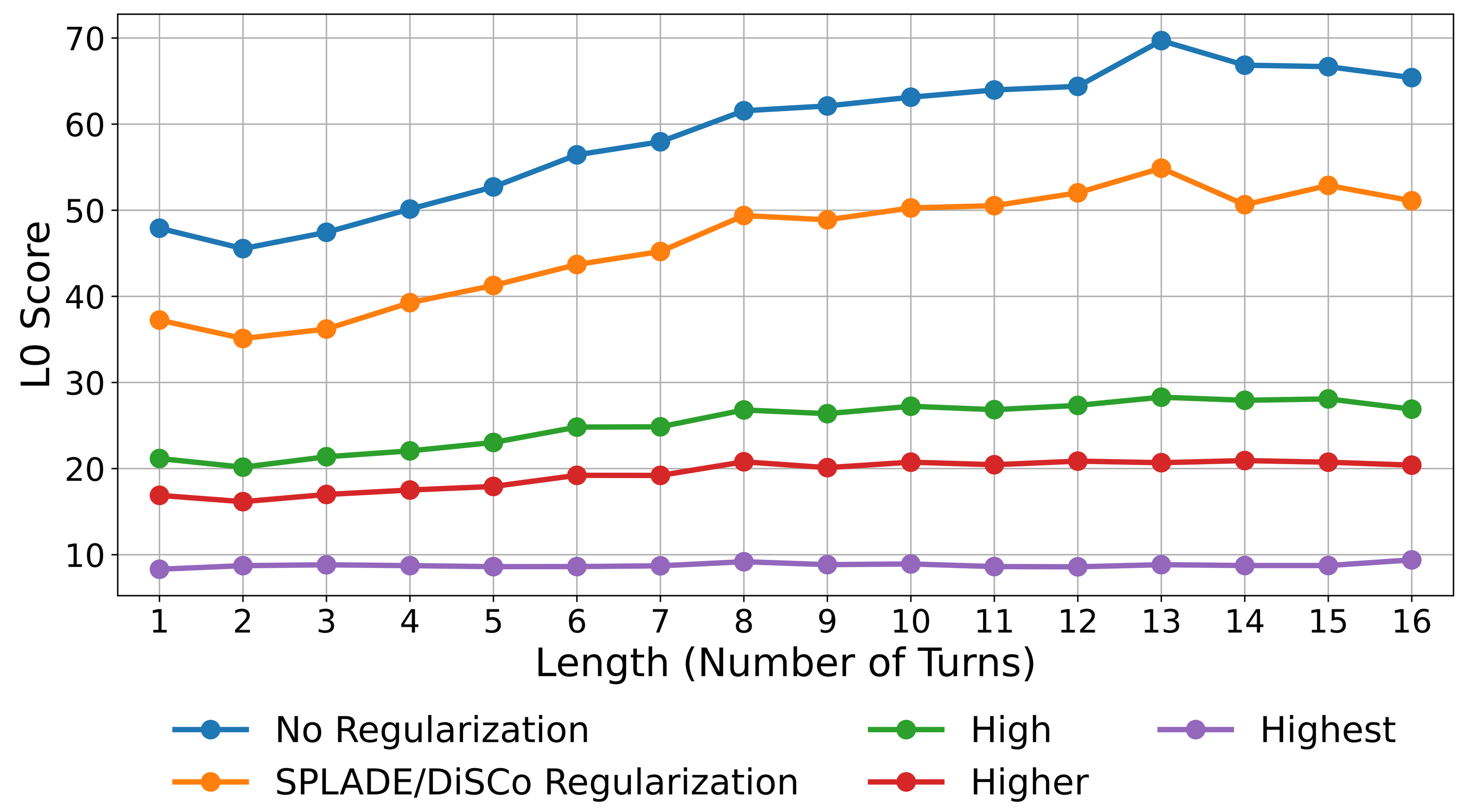

Figure 2. L0 score across conversation lengths for different regularization settings on TopiOCQA. Lower is better.

Increasing regularization substantially reduces the number of activated dimensions, especially for longer conversations. Moderate settings preserve retrieval performance while improving sparsity, indicating that KLD-based distillation can support more efficient sparse representations.

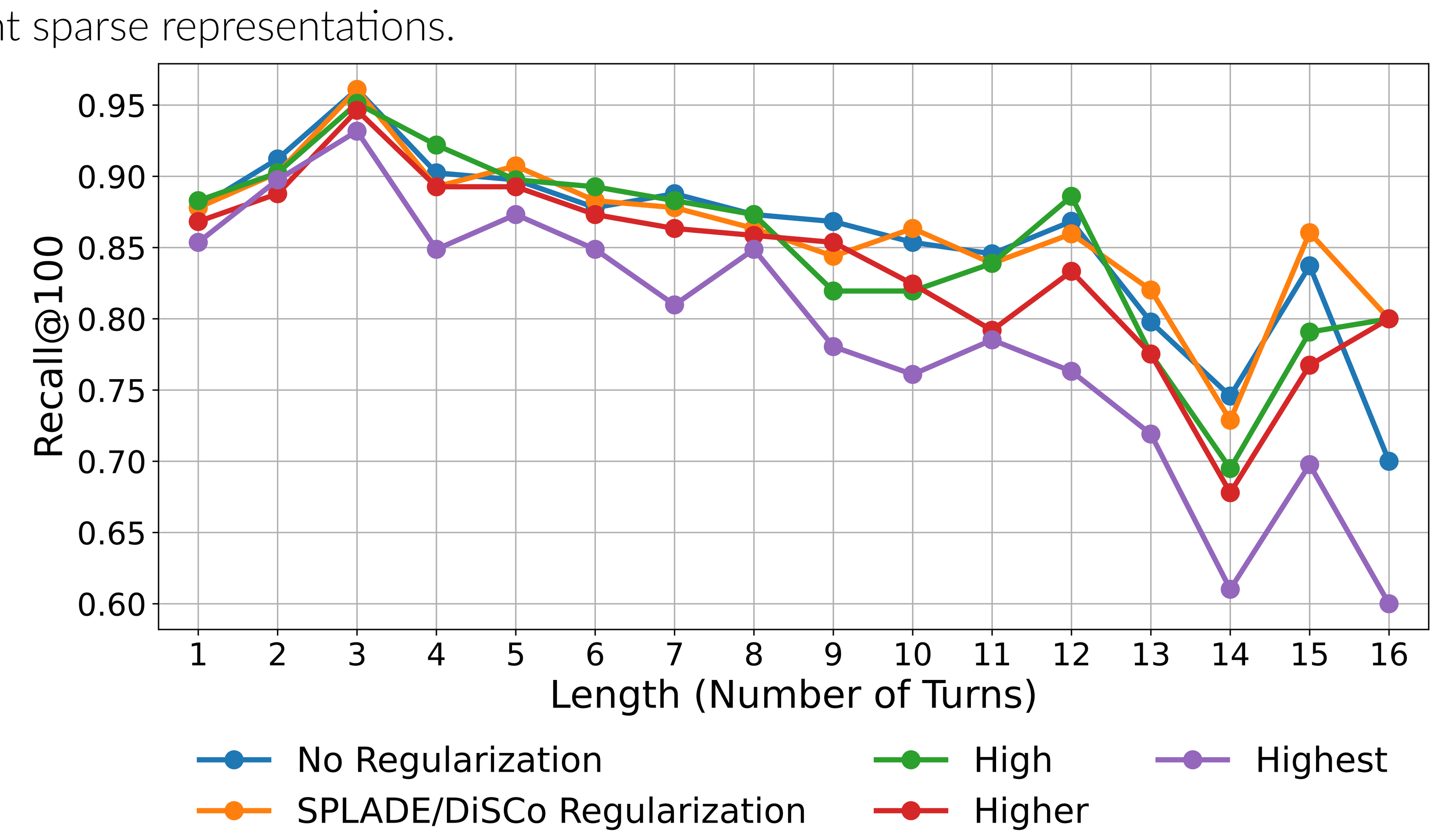

Figure 3. Recall@100 across conversation lengths for different regularization settings on TopiOCQA. Higher is better.

The highest regularization setting achieves the lowest FLOPS but significantly degrades performance. This suggests a useful sparsity–effectiveness trade-off: stronger regularization improves inference efficiency, but excessive sparsification removes retrieval-relevant signal.

Conclusion

- Adding a small contrastive InfoNCE component to the KLD objective improves ranking effectiveness, especially for MRR and nDCG@3.
- Increasing the number of KLD negatives improves the approximation of the teacher distribution, but too many negatives dilute the positive signal and reduce retrieval performance.
- Stronger regularization can substantially improve sparsity and inference efficiency while preserving recall under moderate settings.
- For longer conversations, regularization helps keep sparse representations compact, addressing an important efficiency bottleneck in conversational search.

Takeaways

- Incorporating contrastive learning into KLD improves ranking-oriented retrieval effectiveness.
- The number of KLD negatives should be balanced to preserve the positive training signal.
- Moderate regularization enables more efficient sparse retrieval with limited effectiveness loss.
- Regularization is particularly beneficial for controlling representation size in longer conversations.

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